MANDI VAAZ

A VOICE-FIRST MANDI-PRICE ASSISTANT FOR PUNJAB

Ask about market prices in Punjabi or Punglish. Hear a clear, dated answer in Punjabi.

THE PROBLEM

Market-price information is difficult to act on when access depends on typing, navigating text-heavy services or understanding unfamiliar interfaces. A farmer needs a quick way to ask for a crop and market price in a familiar speaking style.

THE SOLUTION

Mandi Vaaz is a browser-based voice assistant. It transcribes the question, extracts the crop and mandi, checks a local Agmarknet snapshot, and speaks a concise Punjabi reply that includes the record date—or asks for clarification when details are missing.

HOW THE PROTOTYPE WORKS

MIC → FASTAPI → PRISMA STT → ALIAS + FUZZY PARSER → DIALOGUE STATE MACHINE → SQLITE LOOKUP → PUNJABI TEMPLATE → TIMBRE TTS

- **Grounded answers.** Fixed Punjabi templates keep spoken prices tied to database values.
- **Useful dialogue.** Asks for missing details, confirms weak matches and refuses unsupported data.
- **Transparent freshness.** States the source date and flags records older than three days.
- **Graceful fallback.** Can identify a nearby substitute mandi or say when no record is available.

CURRENT SCOPE AND EVIDENCE

Data: 19,959 Agmarknet records (1 Nov 2025–26 May 2026 overall). Only potato, onion and tomato are enabled for answers; useful crop-market records are largely from March/early April 2026, so this is a historical snapshot—not live pricing.

Speech evaluation: 18 read-aloud clips (9 Punjabi, 9 Punglish). Two runs scored 16/18 correct crop+mandi slots (88.9%; 95% Wilson CI 67–97%); a later repeat scored 18/18 on the same clips and code. The earlier failures were valid-but-wrong mandi substitutions, so the perfect repeat is not a 100% accuracy claim.

Engineering checks: 165 Python tests across 10 modules passed in the review environment. The tests include mocked STT/TTS calls; they do not establish field reliability.

LANGUAGE CLAIM — Punjabi and Punglish are the implemented and evaluated paths. Hindi-mixed input is experimental: an earlier 40-clip study scored 11/12 on the Hindi-mixed subset with a candidate pa-IN,hi-IN,en-IN config, but the app's chosen config scored 7/12. The runtime replies remain Punjabi; this is not a full Hindi mode.

WHAT NEEDS TO HAPPEN NEXT

Refresh and validate price data; expand crops and mandi coverage only after checking source quality; add an explicit Hindi input/output mode; require confirmation when the recogniser returns a plausible but potentially wrong mandi; and test with more consented farmer recordings, varied devices and noisy field conditions.